@codedsoul_05

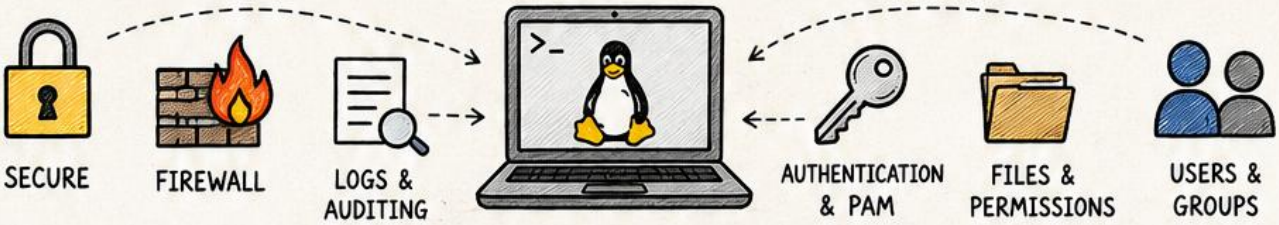
STUDY. REVISE. MASTER.

LINUX SECURITY

15 COMPLETE TOPICS

COMPLETE REVISION NOTES

Concepts | Commands | Diagrams | Best Practices | Interview Q&A



CONCEPTS MADE SIMPLE | EASY TO REVISE | EXAM & INTERVIEW FOCUSED | HIGHER SCORES | REAL WORLD APPLICATIONS

1 Linux Security Fundamentals
- What & Why
- Key Principles
- Security Goals

2 Users, Groups & Permissions
- Users & Groups
- File Permissions
- Special Permissions

3 File Permissions & Ownership
- chmod, chown
- umask
- ACL Basics

4 Network Security & Firewalls
- iptables / nftables
- UFW / Firewalld
- Port & Service Control

5 SSH Security & Hardening
- SSH Config
- Key Auth
- Best Practices

6 Authentication & PAM
- PAM Modules
- Login Security
- Password Policy

7 Sudo & Privilege Management
- Sudo Basics
- Least Privilege
- Sudoers File

8 Service & Process Security
- Secure Services
- systemd Security
- Disable Unused

9 Logs, Auditing & Monitoring
- System Logs
- Auditd
- Monitoring Tools

10 System Hardening & Configuration
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- sysctl Settings
- Secure Configs

11 Malware, Threats & Attack Vecters
- Common Threats
- Malware Types
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12 Vulnerability Management
- Find Vulnerabilities
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13 Backup, Recovery & Incident Response
- Backup Strategies
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14 Common Threats + Defense
- Threats Overview
- Defense in Depth
- Mitigation

15 Quick Revision & Interview Notes
- Key Points
- Commands
- Interview Q&A

SAVE NOW THANK LATER!

"Small steps every day lead to big results."

SECURE TODAY. PROTECT TOMORROW.

LIKE | COMMENT | SAVE | SHARE

1. LINUX FUNDAMENTALS + ARCHITECTURE

WHAT IS LINUX?

- Linux is a free, open-source, and Unix-like operating system.
- Created by Linus Torvalds in 1991.
- It is secure, stable, multi-user, and multi-tasking.
- Used in servers, desktops, embedded systems, cloud, DevOps, and more.
- Linux is not a single OS, but a KERNEL. (Distributions add tools, shells, and apps)

KEY FEATURES

- ✓ Open Source & Free
- ✓ Multi-user & Multi-tasking
- ✓ Portability
- ✓ Security & Stability
- ✓ High Performance
- ✓ Supports a Wide Range of Hardware
- ✓ Customization (Distributions)

LINUX DISTRIBUTIONS (EXAMPLES)

- Debian
- Ubuntu
- Fedora
- CentOS / RHEL
- Arch Linux
- openSUSE
- Kali Linux

TYPES OF LINUX SYSTEMS

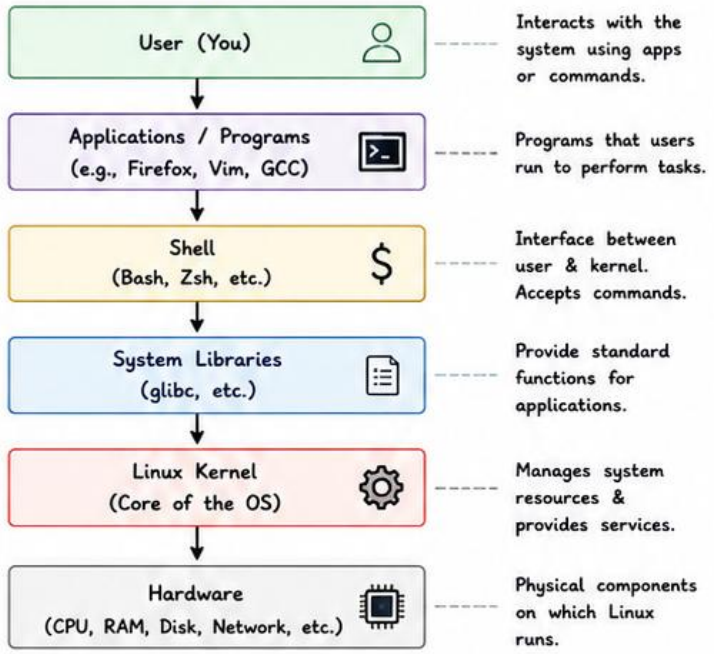
- Desktop Linux → For everyday users (Ubuntu, Fedora)
- Server Linux → For servers (RHEL, CentOS, Debian)
- Embedded Linux → For devices (Raspberry Pi, Router)
- Cloud Linux → For cloud environments (AWS, GCP)

LINUX VS WINDOWS (BASICS)

Feature	Linux	Windows
License	Open Source (Free)	Proprietary (Paid)
Security	More Secure	Less Secure (More Vulnerable)
Users	Multi-user	Single-user
Cost	Free	Paid
Customization	Highly Customizable	Limited
Stability	High	Relatively Low
File System	Ext4, XFS, Btrfs	NTFS, FAT32
Use Case	Servers, DevOps, Cloud, Dev	Gaming, Office, Personal Use

LINUX ARCHITECTURE OVERVIEW

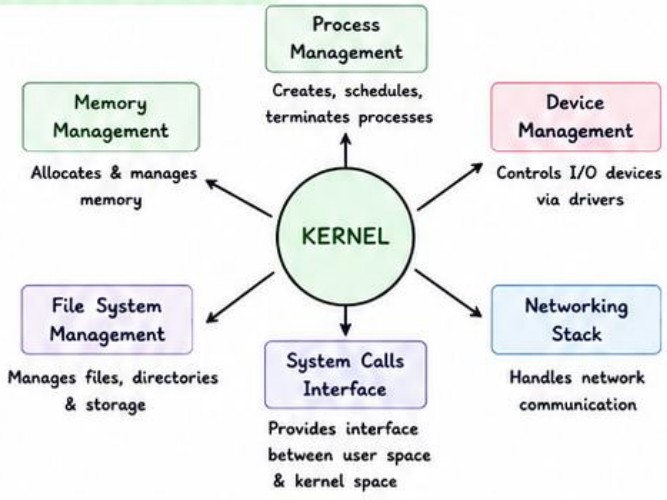
- Linux follows a layered architecture.



LINUX KERNEL – THE HEART

- The kernel is the core component of Linux.
- It manages system resources and acts as an interface between hardware and software.
- Handles: Process Management, Memory Management, Device Management, File Systems, Networking, Security.
- Monolithic kernel with modular design.

KERNEL COMPONENTS



IMPORTANT POINTS

- ✓ Everything in Linux is a file (even devices, processes, sockets).
- ✓ Linux is case-sensitive.
- ✓ Root user (UID 0) has the highest privileges.
- ✓ Standard shells: Bash (most common), Zsh, Fish.
- ✓ Linux follows the philosophy: "Everything is a file, do one thing well, and write programs to work together."

EXAMPLE COMMANDS (BASICS)

ls	→ List files and directories
pwd	→ Print working directory
cd	→ Change directory
cat file	→ View file content
man cmd	→ Manual for a command
uname -a	→ System & kernel information

2. LINUX FILE SYSTEM + DIRECTORY STRUCTURE

WHAT IS LINUX FILE SYSTEM?

- Linux organizes data in a hierarchical tree structure starting from the root directory "/".
- Everything in Linux is a file (files, directories, devices, sockets, etc.)
- Filesystem stores and manages files on storage devices like HDD, SSD, etc.

KEY POINTS

- ✓ Root directory is represented by "/".
- ✓ Linux is case-sensitive.
- ✓ Multiple filesystems can be mounted at different points.
- ✓ Use "df -h" to see disk usage and "du -sh <dir>" to see directory size.

DIRECTORY TREE (OVERVIEW)

- / ← (root)
  - /bin ← Essential user command binaries (ls, cp, mv, cat, etc.)
  - /sbin ← System binaries (admin commands) (fsck, reboot, ifconfig, etc.)
  - /etc ← System configuration files
  - /home ← Home directories of users
  - /root ← Home directory of root user
  - /usr ← User programs and files (most software installed here)
  - /var ← Variable data files (logs, spool, cache, etc.)
  - /tmp ← Temporary files (cleared on reboot)
  - /dev ← Device files (hard drives, USB, terminals, etc.)
  - /proc ← Virtual filesystem for process and system info
  - /sys ← System information (kernel, devices)
  - /lib ← Shared libraries required by binaries in /bin and /sbin
  - /opt ← Optional / third-party software
  - /mnt ← Temporary mount point for filesystems
  - /media ← Mount point for removable media (USB, CD/DVD)
  - /srv ← Data for services provided by the system (web, ftp, etc.)
  - /boot ← Boot loader files, kernel, initrd
  - /lost+found ← Recovered files after filesystem check (fsck)

IMPORTANT DIRECTORIES EXPLAINED

Directory	Purpose	Examples / Contains
/	Root directory, Parent of all directories.	All files and directories exist under /
/bin	Essential user commands	ls, cp, mv, cat, echo, etc.
/sbin	System administration commands	reboot, shutdown, fdisk, ifconfig, fsck, etc.
/etc	Host-specific system configuration	passwd, fstab, hostname, network, ssh, cron, etc.
/home	Home directories of regular users	/home/nikhil, /home/user1
/root	Home directory of root (superuser)	.bashrc, .profile, .ssh
/usr	User installed programs, libraries, docs	/usr/bin, /usr/lib, /usr/share
/var	Files that change frequently	/var/log, /var/cache, /var/mail, /var/spool
/tmp	Temporary files	Temp files (deleted on reboot or manually)
/dev	Device files	sda, sdb, tty, null, zero, etc.
/proc	Process and kernel information	/proc/cpuinfo, /proc/meminfo
/sys	System and kernel information	/sys/class/, /sys/devices/
/lib	Shared libraries	libc.so, libssl.so, ld-linux.so
/opt	Optional add-on software	/opt/google, /opt/docker
/mnt	Mount point for temporary filesystems	Manually mounted partitions
/media	Mount point for removable media	USB, CD/DVD auto-mounted
/srv	Data for services	Web files (/srv/www), FTP files (/srv/ftp)
/boot	Boot related files	vmlinuz, initrd, grub/
/lost+found	Used by fsck to store recovered files	Auto created in ext filesystems

FILESYSTEM HIERARCHY STANDARD (FHS)

- Linux follows FHS (Filesystem Hierarchy Standard).
- It defines what data must be stored in which directory.
- Ensures consistency across different Linux distributions.

USEFUL COMMANDS

`ls /`	→ List root directory contents
`ls /etc`	→ List /etc directory
`cd /home/nikhil`	→ Go to a directory
`pwd`	→ Show current directory
`tree /`	→ Show directory tree (if installed)
`df -h`	→ Disk usage
`du -sh <dir>`	→ Size of a directory

FILE TYPES IN LINUX

-	→ Regular file
d	→ Directory
l	→ Symbolic link
c	→ Character device file
b	→ Block device file
p	→ Named pipe (FIFO)
s	→ Socket

TIPS TO REMEMBER

- ✓ Everything starts from / (root).
- ✓ Linux is case-sensitive (File ≠ file).
- ✓ Do not edit files in /etc unless you know what you are doing.
- ✓ Use absolute path (starts with /) for clarity.
- ✓ Keep /var/log monitored for system logs.

EXAMPLE PATHS

- `/etc/passwd` → Password file
- `/var/log/syslog` → System log file
- `/home/nikhil/docs/file.txt` → File inside user's home
- `/usr/bin/python3` → Python3 binary
- `/dev/sda1` → First partition of first disk

3. LINUX COMMANDS + SHELL BASICS

WHAT IS SHELL?

- Shell is a command-line interpreter.
- It acts as an interface between user and kernel.
- It reads commands from user and executes programs.
- Popular Shells: Bash (default), Zsh, Fish, Sh

TYPES OF SHELL

Shell	Description
sh	Bourne Shell (Original Unix Shell)
bash	Bourne Again SHell (Default in most Linux distros)
zsh	Z Shell (Powerful with auto-completion)
fish	Friendly Interactive SHell (User-friendly)
tcsh	TENEX C Shell (C-like syntax)

BASIC SHELL CONCEPTS

- ✓ Command → An instruction to the system.
- ✓ Options/Flags → Modify command behavior (e.g., -l, -a).
- ✓ Arguments → Additional values (e.g., file name).
- ✓ Path → Location of command or file.
- ✓ Prompt → Indicates shell is ready (e.g., user@host:~$).

SHELL PROMPT

```
nikhil@linux:~$
```

Username: nikhil | Hostname: linux | Current Directory: ~

$ → Normal User
→ Root User

NAVIGATION COMMANDS

Command	Usage	Description	Example
pwd	pwd	Print working directory	pwd
ls	ls [option] [dir]	List files and directories	ls -l /home
cd	cd [dir]	Change directory	cd /etc
cd ..	cd ..	Go to parent directory	cd ..
cd ~	cd ~	Go to home directory	cd ~
cd -	cd -	Go to previous directory	cd -

FILE & DIRECTORY COMMANDS

Command	Usage	Description	Example
touch	touch file.txt	Create an empty file	touch a.txt
mkdir	mkdir dir	Create a directory	mkdir mydir
rm	rm file	Remove a file	rm a.txt
rm -r	rm -r dir	Remove directory (recursive)	rm -r mydir
cp	cp src dest	Copy files/directories	cp a.txt b.txt
mv	mv src dest	Move/Rename	mv a.txt new.txt
cat	cat file	Display file content	cat a.txt
less	less file	View file (scroll)	less a.txt
head	head file	Show first 10 lines	head a.txt
tail	tail file	Show last 10 lines	tail a.txt
tail -f	tail -f file	Follow file output	tail -f syslog

SYSTEM INFORMATION COMMANDS

Command	Usage	Description	Example
uname	uname -a	System information	uname -a
hostname	hostname	Show hostname	hostname
uptime	uptime	System uptime	uptime
df	df -h	Disk free (human readable)	df -h
du	du -sh dir	Directory size	du -sh /var
free	free -h	Memory usage	free -h
top	top	Process overview	top
ps	ps aux	Process snapshot	ps aux
whoami	whoami	Current user	whoami
id	id	User ID and Groups	id
date	date	Show date & time	date
cal	cal	Show calendar	cal

PROCESS MANAGEMENT

Command	Usage	Description	Example
ps	ps aux	List all processes	ps aux
top	top	Live process view	top
kill	kill PID	Kill process by PID	kill 1234
kill -9	kill -9 PID	Force kill process	kill -9 1234
bg	bg [job]	Resume in background	bg %1
fg	fg [job]	Bring to foreground	fg %1
jobs	jobs	List background jobs	jobs

PERMISSION COMMANDS

Command	Usage	Description	Example
chmod	chmod mode file	Change permissions	chmod 755 file
chown	chown user file	Change owner	chown nikhil file
chgrp	chgrp group file	Change group	chgrp dev file
umask	umask	Set default permissions	umask 022

SEARCH COMMANDS

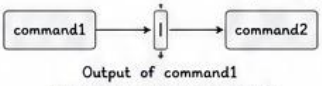
Command	Usage	Description	Example
find	find /path -name file	Find files	find / -name a.txt
grep	grep "text" file	Search text in file	grep "root" /etc/passwd
which	which command	Locate command	which ls
whereis	whereis command	Locate binary, source, man	whereis bash
locate	locate file	Find file (uses db)	locate a.txt

ARCHIVE & COMPRESSION

Command	Usage	Description	Example
tar	tar -cvf file.tar dir	Create tar archive	tar -cvf a.tar dir
tar -xvf	tar -xvf file.tar	Extract tar archive	tar -xvf a.tar
gzip	gzip file	Compress file	gzip a.txt
gunzip	gunzip file.gz	Decompress .gz file	gunzip a.txt.gz
zip	zip file.zip file	Create zip	zip a.zip a.txt
unzip	unzip file.zip	Unzip archive	unzip a.zip

REDIRECTION & PIPES

>	: Redirect output (overwrite)	ls > output.txt
>>	: Redirect output (append)	echo hello >> file.txt
<	: Redirect input	sort < file.txt
|	: Pipe output of one command to another	ps aux | grep nginx



Output of command1 passes as input to command2

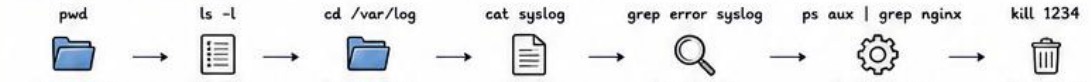
WILDCARDS

*	Matches any number of characters	ls *.txt
?	Matches any single character	ls file?.txt
[abc]	Matches any one of a, b or c	ls file[abc].txt
[a-z]	Matches any character in range	ls file[0-9].txt

TIPS

- ✓ Use Tab key for auto-completion.
- ✓ Use history to see previous commands.
- ✓ Use man <command> to see manual.
- ✓ Use --help with most commands.
- ✓ Practice regularly - that's the key!

EXAMPLE WORKFLOW



REMEMBER

Everything in Linux is a file!

4. USERS + GROUPS + USER MANAGEMENT

WHAT ARE USERS?

- A user is an account that allows a person to access and use the system.
- Each user has a User ID (UID), Primary Group, Home Directory, and Shell.
- Root user (UID 0) has full access and control over the system.

WHAT ARE GROUPS?

- A group is a collection of users.
- Groups help manage permissions for multiple users at once.
- Each group has a Group ID (GID).
- Every user belongs to at least one primary group and can belong to multiple secondary groups.

IMPORTANT FILES

- /etc/passwd → User account info
- /etc/shadow → Encrypted passwords
- /etc/group → Group info
- /etc/gshadow → Secure group info
- /etc/login.defs → Default values for useradd, usermod

USER MANAGEMENT COMMANDS

Command	Description	Example
useradd <user>	Create a new user	useradd nikhil
useradd -m <user>	Create user with home directory	useradd -m nikhil
useradd -d <dir> <user>	Create user with custom home dir	useradd -d /data/nikhil nikhil
useradd -s <shell> <user>	Create user with login shell	useradd -s /bin/bash nikhil
passwd <user>	Set/Change user password	passwd nikhil
usermod <options> <user>	Modify user attributes	usermod -aG sudo nikhil
userdel <user>	Delete a user	userdel nikhil
userdel -r <user>	Delete user and home directory	userdel -r nikhil
id <user>	Show user ID, groups	id nikhil
whoami	Show current logged-in user	whoami
who	Show logged-in users	who

GROUP MANAGEMENT COMMANDS

Command	Description	Example
groupadd <group>	Create a new group	groupadd developers
groupmod <options> <group>	Modify group attributes	groupmod -n devs developers
groupdel <group>	Delete a group	groupdel devs
id <group>	Show group ID and members	id devs
getent group	List all groups	getent group

USERMOD COMMON OPTIONS

Option	Description	Example
-l <newname>	Change username	usermod -l newname oldname
-d <dir>	Change home directory	usermod -d /newhome nikhil
-s <shell>	Change login shell	usermod -s /bin/zsh nikhil
-aG <group>	Add user to supplementary group	usermod -aG sudo nikhil
-G <groups>	Change supplementary groups	usermod -G devs,ops nikhil
-u <uid>	Change user ID	usermod -u 2001 nikhil
-L	Lock user account	usermod -L nikhil
-U	Unlock user account	usermod -U nikhil

VIEW USERS & GROUPS

Command	Description
cat /etc/passwd	View all user accounts
cat /etc/group	View all groups
getent passwd	Show users (from NSS)
getent group	Show groups (from NSS)
groups <user>	Show groups of a user
grep <user> /etc/passwd	Search user in passwd file

USER ACCOUNT INFORMATION (PASSWD FILE FIELDS)

Format: username:x:UID:GID:Full Name:Home Directory:Shell

```
nikhil : x : 1001 : 1001 : Nikhil Rajput : /home/nikhil : /bin/bash
```

- Username
- Password (x means in shadow)
- User ID (UID)
- Group ID (GID)
- Full Name / Comment
- Home Directory
- Login Shell

EXAMPLES – USER MANAGEMENT

```
# Create a user with home directory
useradd -m nikhil
# Set password for user
passwd nikhil
# Add user to 'sudo' group
usermod -aG sudo nikhil
# Change shell to /bin/zsh
usermod -s /bin/zsh nikhil
# Lock and unlock user
usermod -L nikhil     # lock
usermod -U nikhil     # unlock
# Delete user and home directory
userdel -r nikhil
```

EXAMPLES – GROUP MANAGEMENT

```
# Create a group
groupadd devs
# Add user to group
usermod -aG devs nikhil
# See group members
getent group devs
# Rename group
groupmod -n developers devs
# Delete group
groupdel developers
```

DEFAULT SYSTEM GROUPS (EXAMPLES

Group	Purpose
root	Root user group
bin	Binary files group
daemon	System daemons
adm	System administration
sudo	Users allowed to run sudo
users	General users
dev	Device-related access
wheel	(RHEL/CentOS) – sudo users

IMPORTANT NOTES

- ✓ UID 0 is reserved for root user.
- ✓ System users usually have UID 1–999.
- ✓ Regular users start from UID 1000 (varies by distro).
- ✓ Always use strong passwords and follow least privilege principle.
- ✓ Never share root password – use sudo for admin tasks.

BEST PRACTICES

- ✓ Create users with meaningful names.
- ✓ Add users to groups for permission management.
- ✓ Review /etc/passwd and /etc/group regularly.
- ✓ Lock unused accounts.
- ✓ Follow strong password policy.

QUICK REFERENCE

Root User	:	root (UID 0)
Root Group	:	root (GID 0)
Useradd	:	Add user
Usermod	:	Modify user
Userdel	:	Delete user
Groupadd	:	Add group
Groupmod	:	Modify group
Groupdel	:	Delete group

★ USERS & GROUPS HELP YOU MANAGE ACCESS AND SECURITY EFFECTIVELY.

5. FILE PERMISSIONS + OWNERSHIP

WHAT ARE PERMISSIONS?

- Permissions define who can do what with a file or directory.
- Linux is multi-user, so every file has an owner, a group and others with different access rights.
- Permission types:

r	w	x
Read (4)	Write (2)	Execute (1)
View content	Modify content	Run / Access

FILE OWNERSHIP

- Every file/directory has:
  - Owner (User)
  - Group
  - Others (everyone else)
- The owner and group determine permissions.

VIEW PERMISSIONS

`ls -l`

Example:

```
-rw-r--r--  1 nikhil devs 1024 May 10 10:00 file.txt
```

Permissions | Links | Owner | Group | Size | Date | Name

UNDERSTANDING PERMISSION STRING

```
-  rw-  r--  r--
```

File Type | Owner (USR) | Group (GRP) | Others (OTH)

- First character (-, d, l, c, b, p, s) → File type
  - `-` : regular file
  - `d` : directory
  - `l` : symbolic link
  - `c` : character device
  - `b` : block device
  - `p` : named pipe (FIFO)
  - `s` : socket
- Next 9 characters → permissions for Owner, Group, Others.
- Order: rwx (Read, Write, Execute)

PERMISSION VALUES (NUMERIC)

Permission	Symbol	Value	Meaning
Read	r	4	View file contents / list directory
Write	w	2	Modify file / create, delete in directory
Execute	x	1	Run file / access directory

CALCULATE PERMISSIONS

r	w	x	Total	Permission
4	2	4	7	→ rwx (read, write, execute)
4	2	0	6	→ rw- (read, write)
4	0	1	5	→ r-x (read, execute)
4	0	0	4	→ r-- (read only)
0	2	1	3	→ -wx (write, execute)
0	2	0	2	→ -w- (write only)
0	0	1	1	→ --x (execute only)
0	0	0	0	→ --- (no permission)

EXAMPLE: NUMERIC PERMISSIONS

Numeric	Owner	Group	Others	Symbolic	Meaning
755	r w x	r - x	r - x	-rwxr-xr-x	Owner: rwx Group: r-x Others: r-x
644	r w -	r - -	r - -	-rw-r--r--	Owner: rw- Group: r-- Others: r--
600	r w -	- - -	- - -	-rw-------	Owner: rw- Group: --- Others: ---
777	r w x	r w x	r w x	-rwxrwxrwx	All have full access
700	r w x	- - -	- - -	-rwx------	Owner full access

IMPORTANT COMMANDS

chmod (Change Permissions)

```
chmod [options] mode file/directory
```

Examples:

```
chmod 755 file.txt     # set 755
chmod u+x script.sh    # add execute for owner
chmod g-w file.txt     # remove write for group
```

chown (Change Owner)

```
chown [options] user file/directory
```

Examples:

```
chown nikhil file.txt     # change owner
chown -R nikhil /data     # recursive
```

chgrp (Change Group)

```
chgrp [options] group file/directory
```

Examples:

```
chgrp devs file.txt     # change group
chgrp -R devs /data     # recursive
```

own & group together

```
chown user:group file/directory
```

Example:

```
chown nikhil:devs file.txt
chown -R nikhil:devs /data
```

chmod MODES

1) Symbolic Mode (using letters)

Who	Letters	Meaning
u	user (owner)	File owner
g	group	File group
o	others	Others (everyone else)
a	all	All (u + g + o)

Operators

+	add permission
-	remove permission
=	set exact permission

Examples:

```
chmod u+r file.txt    # add read for owner
chmod g-w file.txt    # remove write for group
chmod o+x file.txt    # add execute for others
chmod a=r file.txt    # set read for everyone only
```

chmod NUMERIC MODE

Set permissions using 3 digits (owner-group-others)

Example:

```
chmod  754  file.txt
       |    |
       7    5    4
       |    |    |
     owner group others
```

7 (rwx) for owner
5 (r-x) for group
4 (r--) for others

SPECIAL PERMISSIONS

Permission	Symbol	Octal	Purpose
Setuid	u+s	4000	Run file as owner's user
Setgid	g+s	2000	Run file as group's user (or inherit group on dir)
Sticky Bit	o+t	1000	Only owner can delete files in directory

Example:

```
chmod 4755 program    # setuid
chmod 2775 dir        # setgid on directory
chmod 1777 /tmp       # sticky bit on /tmp
```

DIRECTORY PERMISSIONS

- For files: r = read content, w = modify, x = execute
- For directories:
  - r : list files in directory
  - w : create/delete files in directory
  - x : enter/access directory (cd)

Example:

```
-rwxr-x---  file.txt
drwxrwxr-x  mydir/
```

To access a file inside a directory, you need execute (x) permission on the directory.

CHECK ACCESS

→ test Permission

```
-r file   # check read
-w file   # check write
-x file   # check execute
[ -r file ] && echo "Readable"
```

umask (Default Permission)

umask sets default permission for new files/dirs.

```
umask          # view current umask
umask 022      # files: 644, dirs: 755 (default)
umask 077      # files: 600, dirs: 700
```

BEST PRACTICES

- ✓ Give minimum permissions required.
- ✓ Do not use 777 on files/directories.
- ✓ Use groups to manage access, not multiple users.
- ✓ Regularly review ownership and permissions.
- ✓ Use strong umask settings.

QUICK RECAP

- Every file has: Owner, Group, Others
- Permissions: r (4), w (2), x (1)
- View: ls -l
- Change: chmod, chown, chgrp
- Good permissions = Good security

EXAMPLE WORKFLOW

1. Create file: `touch file.txt` →
2. Check: `ls -l file.txt` →
3. Change owner: `chown nikhil file.txt` →
4. Change group: `chgrp devs file.txt` →
5. Set permission: `chmod 640 file.txt` →
6. Verify: `ls -l file.txt`

6. PROCESSES + SERVICES + SYSTEM MANAGEMENT

WHAT ARE PROCESSES?

- A process is a running instance of a program.
- Each process has a PID (Process ID).
- Linux is a multi-tasking OS that can run many processes simultaneously.
- We can view, manage, kill and monitor processes.

WHAT ARE SERVICES?

- A service (or daemon) is a background process that starts at boot and runs continuously.
- Examples: sshd, apache2, mysql, cron, etc.
- Managed using systemd in modern Linux.

SYSTEM MANAGEMENT

- System management includes tasks to monitor, configure and maintain the system.
- Includes: processes, services, resources, logs, startup, scheduling, backups, etc.

PROCESS MANAGEMENT COMMANDS

Command	Usage	Description	Example
ps	ps aux	Show all running processes	ps aux
ps	ps -ef	Show processes in full format	ps -ef
top	top	Real-time process monitor	top
htop	htop	Interactive process viewer	htop
pgrep	pgrep <name>	Find PID of a process	pgrep nginx
pidof	pidof <name>	Find PID of a process	pidof sshd
kill	kill <PID>	Terminate a process (SIGTERM)	kill 1234
kill -9	kill -9 <PID>	Force kill a process (SIGKILL)	kill -9 1234
nice	nice -n <val> cmd	Start process with priority	nice -n 10 cmd
renice	renice <val> -p <PID>	Change priority of running process	renice 10 -p 1234
pstree	pstree	Show processes in tree view	pstree

ps aux OUTPUT EXPLAINED

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
User	Process ID	CPU Usage	Memory Usage	Virtual Memory Size	Resident Set Size	Terminal	Process State	Start Time	CPU Time	Command Name

STAT (Process State)

R	Running
S	Sleeping
D	Uninterruptible Sleep (usually I/O)
T	Stopped / Traced
Z	Zombie (defunct)

NICE VALUES (Priority)

-20 (highest priority)
0 (default priority)
19 (lowest priority)

SERVICE MANAGEMENT (systemctl)

Command	Description	Example
systemctl start <service>	Start a service	systemctl start sshd
systemctl stop <service>	Stop a service	systemctl stop nginx
systemctl restart <service>	Restart a service	systemctl restart apache2
systemctl reload <service>	Reload a service (re-read config)	systemctl reload sshd
systemctl enable <service>	Enable service at boot	systemctl enable nginx
systemctl disable <service>	Disable service at boot	systemctl disable nginx
systemctl status <service>	Check service status	systemctl status sshd
systemctl is-active <service>	Check if service is active	systemctl is-active sshd
systemctl list-units --type=service	List all services	systemctl list-units -type=service

COMMON SERVICE STATES

active (running)	→	Service is running
inactive (dead)	→	Service is stopped
failed	→	Service failed to start
enabled	→	Service starts at boot
disabled	→	Service not started at boot

SYSTEM RESOURCE & MONITORING COMMANDS

Command	Description	Example
uptime	Show system uptime and load	uptime
free -h	Show memory usage (human readable)	free -h
df -h	Show disk usage	df -h
du -sh <dir>	Show size of directory	du -sh /var
vmstat	Report virtual memory statistics	vmstat 1
iostat	CPU / I-O statistics	iostat -x 1
sar	System Activity Reporter	sar -u 1 5
watch <cmd>	Run command periodically	watch -n 2 df -h

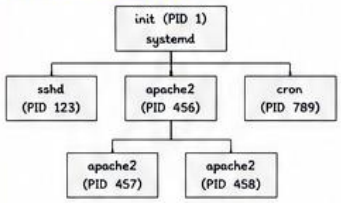
BOOT & RUNLEVELS (systemd TARGETS)

Command	Description	Example
systemctl get-default	Show default target	systemctl get-default
systemctl set-default <target>	Set default target	systemctl set-default multi-user.target
systemctl list-units --type=target	List all targets	systemctl list-units --type=target

COMMON TARGETS

multi-user.target	→	Text mode (default runlevel 3)
graphical.target	→	Graphical mode (runlevel 5)
rescue.target	→	Single user mode
poweroff.target	→	Power off
reboot.target	→	Reboot

VIEW PROCESS TREE



CHANGE PROCESS PRIORITY

Lower nice value = higher priority



MANAGE STARTUP PROGRAMS

Method	Description	Example
systemctl enable	Enable a service (start at boot)	systemctl enable nginx
systemctl disable	Disable a service (do not start at boot)	systemctl disable nginx
chkconfig (old SysV)	Manage services (SysV systems)	chkconfig --list
update-rc.d (Debian)	Manage init scripts	update-rc.d -f apache2 defaults

VIEW LOGS

Log File	Description
/var/log/syslog	General system logs
/var/log/messages	System messages
/var/log/auth.log	Authentication logs
/var/log/kern.log	Kernel logs
/var/log/dmesg	Boot messages
journalctl	View systemd logs

Examples:

```
journalctl -xe          # View logs with errors
journalctl -u sshd      # Logs for a service
```

SCHEDULE TASKS (CRON)

Command	Description	Example
crontab -e	Edit current user's cron jobs	crontab -e
crontab -l	List user's cron jobs	crontab -l
crontab -r	Remove user's cron jobs	crontab -r
/etc/crontab	System-wide cron file	⁰/etc/crontab
/etc/cron.d/	Additional cron jobs	/etc/cron.d/
/etc/cron.daily/	Scripts run daily	ls /etc/cron.daily/

CRON FORMAT

```
*  *  *  *  *   command_to_execute
↑  ↑  ↑  ↑  ↑
Min  Hour  Dom  Month  Day (0-6)
(0-59)  (0-23)  (1-31)  (1-12)  (0=Sun)
```

MANAGE USERS & SYSTEM

Command	Description	Example
who	Who is logged in	who
w	Detailed who information	w
last	Last logged in users	last
hostnamectl	Show / set hostname	hostnamectl
timedatectl	Date, time and timezone	timedatectl
shutdown -h now	Shut down system	shutdown -h now
reboot	Reboot system	reboot

BEST PRACTICES

- ✓ Regularly monitor processes and resource usage.
- ✓ Disable unnecessary services.
- ✓ Keep system and packages up to date.
- ✓ Enable only required services at boot.
- ✓ Check logs regularly for issues.
- ✓ Use appropriate permissions for all services and files.
- ✓ Use systemctl, not old service scripts.
- ✓ Automate tasks with cron wisely.

7. NETWORKING + LINUX NETWORK COMMANDS

WHAT IS NETWORKING?

- Networking allows systems to communicate with each other over a network.
- Linux provides powerful tools to configure, test, monitor and troubleshoot networks.



IMPORTANT FILES

- /etc/hosts → Hostname to IP mapping
- /etc/resolv.conf → DNS configuration
- /etc/network/interfaces → Debian/Ubuntu config
- /etc/sysconfig/network-scripts/ → RHEL/CentOS config
- /etc/hostname → System hostname

NETWORK INTERFACES

- An interface is a connection point for sending/receiving data.
- Common interface names: lo (loopback), eth0, ens33, enp0s3, wlan0
- View interfaces using: ip a or ifconfig

NETWORK CONFIGURATION COMMANDS

Command	Description	Example
ip addr or ip a	Show / configure IP addresses	ip a
ip link	Show / set interface status	ip link
ip link set <iface> up	Bring interface up	ip link set ens33 up
ip link set <iface> down	Bring interface down	ip link set ens33 down
ifconfig	Display / configure interfaces (legacy)	ifconfig
hostname	Show or set system hostname	hostname
hostnamectl set-hostname <name>	Set system hostname	hostnamectl set-hostname server1
nmcli device status	Show NetworkManager status	nmcli device status
nmcli con show	Show network connections	nmcli con show

CHECK CONNECTIVITY

Command	Description	Example
ping <host>	Test connectivity to host	ping 8.8.8.8
ping -c 4 <host>	Send 4 packets and exit	ping -c 4 google.com
traceroute <host>	Trace route to host	traceroute google.com
mtr <host>	Network diagnostic tool (ping + traceroute)	mtr google.com
ss -tuln	Show listening ports (TCP/UDP)	ss -tuln
netstat -tuln	Show listening ports (legacy)	netstat -tuln
telnet <host> <port>	Test port connectivity	telnet 8.8.8.8 53
nc -zv <host> <port>	Check if port is open	nc -zv google.com 80

IP ADDRESS INFORMATION

Command	Description	Example
ip addr show	Show all IP addresses	ip addr show
ip -4 addr show	Show IPv4 addresses	ip -4 addr show
ip -6 addr show	Show IPv6 addresses	ip -6 addr show
ip addr show <iface>	Show IP of specific interface	ip addr show ens33

ROUTING COMMANDS

Command	Description	Example
ip route show	Show routing table	ip route show
ip route add <net> via <gw>	Add static route	ip route add 10.0.0.0/24 via 192.168.1.1
ip route del <net>	Delete static route	ip route ddl 10.0.0.0/24
ip route add default via <gw>	Set default gateway	ip route add default via 192.168.1.1

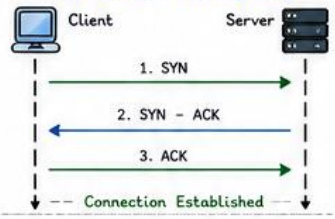
DNS & NAME RESOLUTION

Command	Description	Example
cat /etc/resolv.conf	View DNS configuration	cat /etc/resolv.conf
nslookup <host>	Query DNS (legacy)	nslookup google.com
dig <host>	DNS lookup (detailed)	dig google.com
host <host>	Simple DNS lookup	host google.com

VIEW OPEN PORTS & CONNECTIONS

Command	Description	Example
ss -tuln	Listening TCP/UDP ports	ss -tuln
ss -antp	All TCP connections with process	ss -antp
lsof -i	List open network files/ports	lsof -i
lsof -i :80	Who is using port 80	lsof -i :80

TCP 3-WAY HANDSHAKE



COMMON NETWORK TOOLS

Tool	Description	Example
ip	Modern network configuration tool	ip a, ip route
ss	Socket statistics (faster than netstat)	ss -tuln
nmcli	NetworkManager CLI	nmcli device status
ethtool	View / change ethernet settings	ethtool eth0
iwconfig	Wireless configuration (legacy)	iwconfig wlan0
tcpdump	Packet analyzer	tcpdump -i eth0
wireshark	Network protocol analyzer (GUI)	wireshark

FIREWALL COMMANDS (firewalld)

Command	Description	Example
systemctl status firewalld	Check firewall status	systemctl status firewalld
firewall-cmd --list-all	List all firewall rules	firewall-cmd --list-all
firewall-cmd --list-ports	List allowed ports	firewall-cmd --list-ports
firewall-cmd --add-port=80/tcp --permanent	Open port permanently	firewall-cmd --add-port=80/tcp --permanent
firewall-cmd --reload	Reload firewall	firewall-cmd --reload

LOOPBACK INTERFACE

- The loopback interface is used for communication within the same system.
- It has the IP: 127.0.0.1
- Interface: lo
- Example: curl http://127.0.0.1



PRACTICAL EXAMPLES

```
# 1. Check IP address
ip a

# 2. Bring interface up
sudo ip link set ens33 up

# 3. Set static IP (temporary)
sudo ip addr add 192.168.1.100/24 dev ens33

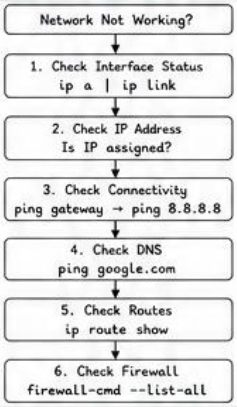
# 4. Add default gateway
sudo ip route add default via 192.168.1.1

# 5. Test connectivity
ping -c 4 8.8.8.8

# 6. Check DNS
cat /etc/resolv.conf

# 7. Check listening ports
ss -tuln
```

TROUBLESHOOTING FLOW



IMPORTANT NOTES

- ✓ Use "ip" and "ss" – modern and fast.
- ✓ ifconfig and netstat are legacy.
- ✓ Always test connectivity layer by layer.
- ✓ Keep /etc/hosts updated for local mappings.
- ✓ Use firewall to allow only required ports.

QUICK REFERENCE

View IP	ip a
View Routes	ip route show
Ping Host	ping <host>
DNS Lookup	dig <host>
Listening Ports	ss -tuln
Open Port (firewalld)	firewall-cmd --add-port=80/tcp --permanent

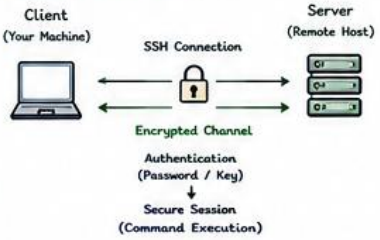
Network knowledge is power – connect, configure, secure!

8. SSH + SECURE REMOTE ACCESS

WHAT IS SSH?

- SSH (Secure Shell) is a cryptographic network protocol for secure remote access.
- It provides secure login, command execution, and file transfer over an insecure network.
- Default port: 22
- It replaces insecure protocols like Telnet, FTP, rlogin, rsh.

HOW SSH WORKS



KEY FEATURES

- ✓ Confidentiality – all data is encrypted
- ✓ Integrity – data cannot be altered
- ✓ Authentication – verify user/server identity
- ✓ Authorization – control what user can do
- ✓ Tunneling – secure other protocols
- ✓ Port Forwarding – access internal services
- ✓ File Transfer – scp, sftp

SSH BASIC COMMANDS

Command	Description	Example
ssh user@host	Connect to remote host	ssh nikhil@192.168.1.10
ssh -p port user@host	Connect using custom port	ssh -p 2222 nikhil@host
ssh -i key user@host	Connect using private key	ssh -i id_rsa nikhil@host
exit / logout	Close SSH session	exit
who	Show logged in users	who
uptime	Show system uptime	uptime
hostname	Show system hostname	hostname

AUTHENTICATION METHODS

Password Authentication

- User enters password to login.
- Easy to use but less secure.
- Avoid using root login.

Key-based Authentication

- Uses public/private key pair.
- More secure than passwords.
- Recommended for production.

SSH KEY PAIR (PUBLIC / PRIVATE)

- Private Key: kept secret on your machine.
- Public Key: placed on the remote server (~/.ssh/authorized_keys).
- Works on asymmetric cryptography.
- No password required once setup.



GENERATE SSH KEY

```
# Generate RSA key (recommended)
ssh-keygen -t rsa -b 4096 -C "your_email@domain.com"

# Default location:
~/.ssh/id_rsa        (private key)
~/.ssh/id_rsa.pub    (public key)

# Press Enter to accept default path
# Optionally set a passphrase for extra security
```

COPY PUBLIC KEY TO SERVER

- Method 1: Using ssh-copy-id (easiest)

```
ssh-copy-id user@host
```

- Method 2: Manual

1. cat ~/.ssh/id_rsa.pub
2. Copy the output
3. On server, add to ~/.ssh/authorized_keys
4. Set permissions:

```
mkdir -p ~/.ssh
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
```

FILE TRANSFER OVER SSH

SCP (Secure Copy)

Copy files between local and remote.

```
# Copy file to remote
scp file.txt user@host:/path/

# Copy file from remote
scp user@host:/path/file.txt ./

# Copy directory recursively
scp -r mydir user@host:/path/
```

SFTP (Secure File Transfer Protocol)

Interactive file transfer.

```
sftp user@host
```

```
sftp> ls
sftp> get file.txt
sftp> gut file.txt
sftp> bye
```

TUNNELING & PORT FORWARDING

Local Port Forwarding

(Access remote service locally)

Localhost:8080 → Remote:80

```
ssh -L 8080:localhost:80 user@host
```

Now access http://localhost:8080 (traffic goes to remote:80)

Remote Port Forwarding

(Expose local service to remote)

Remote:9090 → Localhost:3000

```
ssh -R 9090:localhost:3000 user@host
```

Remote can access your local service on port 3000 via 9090

Dynamic Port Forwarding (SOCKS Proxy)

```
ssh -D 1080 user@host
```

Use localhost:1080 in your browser/proxy settings.

SSH CONFIG FILE

File: ~/.ssh/config

Simplify SSH connections.

```
Host myserver
    HostName 192.168.1.10
    User nikhil
    Port 2222
    IdentityFile ~/.ssh/id_rsa
```

Now connect using:

```
ssh myserver
```

SECURE SSH BEST PRACTICES

- ✓ Use key-based authentication, disable passwords.
- ✓ Change default SSH port (not 22).
- ✓ Disable root login.
- ✓ Use a strong passphrase for private key.
- ✓ Keep SSH server updated.
- ✓ Use Fail2Ban to prevent brute force attacks.
- ✓ Limit user access (AllowUsers in config).

IMPORTANT SSH SERVER CONFIG (/etc/ssh/sshd_config)

Directive	Recommended Setting	Description
Port 2222	Change default port	
PermitRootLogin no	Disable root login	
PasswordAuthentication no	Disable password login	
PubkeyAuthentication yes	Allow key-based auth	
AllowUsers nikhil	Allow only specific users	
MaxAuthTries 3 1	Limit authentication attempts	
Banner /etc/issue.net	Show warning banner	

After Changes

1. sudo systemctl restart sshd
2. Test connection before closing current session!

TROUBLESHOOTING

- Connection refused: Server down / wrong IP / port blocked.
- Permission denied: Wrong user / password / key.
- Too many authentication failures: wait and try again.
- Use verbose mode:

```
ssh -vvv user@host
```

COMMON SSH FILES & LOCATIONS

File / Location	Purpose
~/.ssh/id_rsa	Private key
~/.ssh/id_rsa.pub	Public key
~/.ssh/authorized_keys	Authorized public keys
~/.ssh/known_hosts	Stores known hosts
/etc/ssh/sshd_config	SSH server configuration
/var/log/auth.log	SSH logs (Ubuntu/Debian)

QUICK RECAP

- SSH = Secure remote access
- Use keys, not passwords
- scp / sftp for file transfer
- Port forwarding for secure tunneling
- Follow best practices to stay secure

SSH makes remote access secure, encrypted, and reliable. Use it smartly & stay secure!

9. LINUX SECURITY FUNDAMENTALS

WHAT IS LINUX SECURITY?

- Linux security means protecting the system, data and services from unauthorized access, modification, disclosure or destruction.
- It is a combination of:
  - ✓ Strong user & access control
  - ✓ Least privilege principle
  - ✓ Secure configuration
  - ✓ Auditing & monitoring
  - ✓ Timely updates & patching

GOALS OF LINUX SECURITY

- ✓ Confidentiality – ensure data is accessible only to authorized users.
- ✓ Integrity – protect data from unauthorized changes.
- ✓ Availability – ensure systems and services are available when needed.
- ✓ Accountability – actions of users can be traced and monitored.

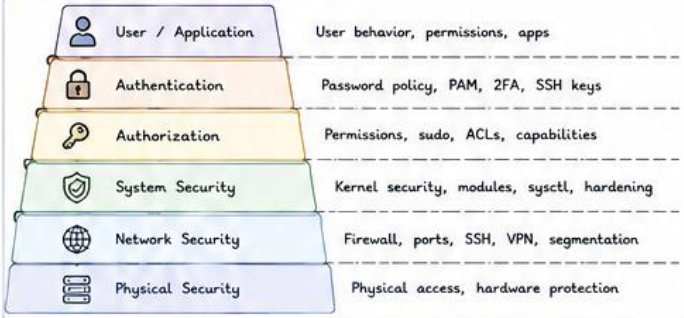
KEY PRINCIPLES

1. Least Privilege – give only required access.
2. Defense in Depth – multiple layers of security.
3. Fail Secure – system should fail safely.
4. Keep it Simple – avoid unnecessary services and complexity.
5. Stay Updated – apply security patches regularly.

COMMON THREATS IN LINUX

- Unauthorized Access – weak passwords, default accounts, exposed services.
- Malware & Backdoors – viruses, trojans, rootkits.
- Privilege Escalation – gaining higher privileges from low-privileged user.
- Network Attacks – sniffing, spoofing, brute force, DoS/DDoS.
- Misconfiguration – open ports, insecure permissions, unnecessary services.
- Insider Threats – misuse of legitimate access.
- Software Vulnerabilities – outdated packages, unpatched flaws.

SECURITY LAYERS IN LINUX



IMPORTANT SECURITY CONCEPTS

Authentication
Verifying the identity of a user or process.

Authorization
Deciding what an authenticated user is allowed to do.

Accounting / Auditing
Recording user activities for accountability.

Non-repudiation
Ensuring actions cannot be denied later.

Least Functionality
Run only necessary services and tools.

BASIC SECURITY BEST PRACTICES

1. Use strong passwords and enforce password policy.
2. Disable root login and use sudo for admin tasks.
3. Use SSH keys instead of passwords.
4. Keep system, packages and firmware updated.
5. Disable unnecessary services and ports.
6. Configure firewall to allow only required traffic.
7. Set proper file permissions and ownership.
8. Monitor logs and system activity regularly.
9. Backup important data regularly.
10. Follow the principle of least privilege.

IMPORTANT FILES & DIRECTORIES

Path	Purpose
/etc/passwd	User account information
/etc/shadow	Encrypted user passwords
/etc/group	Group information
/etc/sudoers	sudo access configuration
/etc/ssh/sshd_config	SSH server configuration
/etc/hosts.allow	Hosts allowed (legacy)
/etc/hosts.deny	Hosts denied (legacy)
/var/log/	System log files
/etc/cron.d/	Scheduled tasks
/etc/fstab	File system mount config

COMMON SECURITY RISKS & MITIGATION

Risk	Description	Mitigation
Weak Passwords	Easy to guess or brute force	Use strong passwords, policy, 2FA
Root Login via SSH	Direct root login is dangerous	Disable root login, use sudo
Open Unnecessary Ports	Attack surface increases	Close unused ports, use firewall
Unpatched System	Known vulnerabilities exploited	Regular updates & patching
Insecure Permissions	Sensitive files accessible	Set proper permissions & ownership
Exposed Services	Unneeded services running	Disable unnecessary services
Lack of Monitoring	Attacks go undetected	Monitor logs, use alerts
Physical Access	System can be tampered	Restrict physical access

COMMON SECURITY TOOLS & UTILITIES

Tool / Command	Purpose	Example
passwd	Change user password	passwd username
chage	Password aging policy	chage -l username
sudo	Run commands as another user	sudo command
sshd	SSH server	systemctl status sshd
ufw / iptables	Firewall	ufw enable
fail2ban	Block brute force attacks	fail2ban-client status
auditd	Auditing & logging	systemctl status auditd
last / who	Login history	last
chkrootkit	Rootkit scanner	chkrootkit
lynis	Security auditing tool	lynis audit system

EXAMPLE: BASIC SECURITY COMMANDS

```
# Check user id
id

# Check last logins
last

# Check listening ports
ss -tuln

# Check open ports with firewall
ufw status

# Check failed login attempts
grep "Failed password" /var/log/auth.log

# Check sudo users
cat /etc/sudoers

# Check permissions
ls -l /etc/passwd

# Check running services
systemctl list-units --type=service
--state=running
```

SYSTEM HARDENING CHECKLIST

- ✓ Disable root login via SSH
- ✓ Use SSH key authentication
- ✓ Enforce strong password policy
- ✓ Keep system and packages updated
- ✓ Enable firewall and configure rules
- ✓ Disable unnecessary services
- ✓ Set proper file permissions
- ✓ Enable logging and monitoring
- ✓ Review cron jobs and startup scripts
- ✓ Backup critical data

QUICK REFERENCE

Task	Command
View IP address	ip a
View routes	ip route
Check Disk Usage	df -h
Check Memory Usage	free -h
View Logs	journalctl -xe
View SSH Status	systemctl status sshd
Restart SSH	systemctl restart sshd
Check Kernel	uname -r

Security is not a product, but a process. Stay alert, stay updated, stay secure!

10. AUTHENTICATION + PAM

WHAT IS AUTHENTICATION?

- Authentication verifies the identity of a user or process.
- It answers the question: "Who are you?"
- Linux uses multiple methods to authenticate users and services.
- Pluggable Authentication Modules (PAM) provide a flexible framework.

WHAT IS PAM?

- PAM (Pluggable Authentication Modules) is a framework that allows system administrators to configure authentication, authorization, password management and session management.
- Applications use PAM to perform security-related tasks.

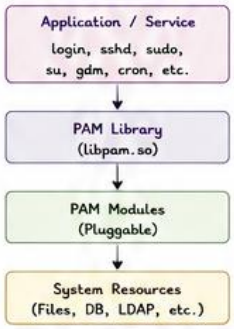
PAM FEATURES

- ✓ **Modular** – add or remove modules easily
- ✓ **Reusable** – one configuration for many services
- ✓ **Centralized** – consistent security policy
- ✓ **Extensible** – supports new auth methods
- ✓ **Ordered Control** – fine-grained control of module execution

AUTHENTICATION METHODS IN LINUX

Method	Description	Used By / Example
Password Authentication	Verify using username & password	login, ssh, sudo
Public Key Authentication	Verify using SSH keys	ssh
Kerberos Authentication	Network authentication using tickets	enterprise environments
Smart Card / Token	Hardware token / smart card	2FA, enterprise
Biometric Authentication	Fingerprint, face, etc.	Some desktop systems
MFA (Multi-Factor)	Combination of two or more factors	SSH + OTP, 2FA, etc.

PAM ARCHITECTURE



1. Application calls PAM.
2. PAM reads configuration from /etc/pam.d/<service>.
3. PAM loads modules in the defined order.
4. Modules perform tasks:
   - Authentication
   - Account management
   - Password management
   - Session management
5. PAM returns success or failure to the application.

PAM STACK TYPES (FACILITIES)

Type	Purpose	Example Services	Description
auth	Authentication	login, sshd, sudo	Verify user identity.
account	Account Management	login, sshd, sudo	Check if user account is valid (expiry, restrictions, etc.)
password	Password Management	passwd, chpasswd	Change or validate user passwords.
session	Session Management	login, sshd, su	Manage user session (environment, limits, logging, etc.).

PAM CONFIGURATION FILES

Path	Purpose
/etc/pam.d/<service>	PAM configuration for a specific service (e.g., login, sshd)
/etc/pam.d/system-auth	Common authentication configuration (RHEL/CentOS)
/etc/pam.d/password-auth	Common password configuration (RHEL/CentOS)
/etc/pam.d/common-auth	Common auth (Debian/Ubuntu)
/etc/pam.d/common-account	Common account (Debian/Ubuntu)
/etc/pam.d/common-password	Common password (Debian/Ubuntu)
/etc/pam.d/common-session	Common session (Debian/Ubuntu)

Actual files used depend on the distribution.

COMMON PAM MODULES

Module	Purpose
pam_unix.so	Local system authentication (e.g., /etc/shadow)
pam_sss.so	Authenticate against SSSD (LDAP, AD, IPA, etc.)
pam_ldap.so	Authenticate against LDAP
pam_krb5.so	Kerberos v5 authentication
pam_faillock.so	Failed login attempts tracking
pam_securetty.so	Restrict root login to secure terminals
pam_limits.so	Set resource limits (ulimits)
pam_env.so	Set environment variables
pam_motd.so	Message of the day
pam_lastlog.so	Print last login information
pam_tally2.so	Legacy failed login tracking (obsolete)

PAM CONTROL FLAGS (HOW MODULE RESULTS ARE HANDLED)

Flag	Meaning	Description
required	Must succeed	If it fails, overall result is failure, but other modules still run.
requisite	Must succeed	If it fails, authentication fails immediately.
sufficient	May succeed	If it succeeds and no prior required failed, authentication is successful, others are skipped.
optional	Optional	If it fails, it does not affect the overall result.
[success=n default=ignore]	Custom actions	Jump to module n on success, ignore on failure/other result.
[default=die]	Custom actions	Fail immediately for any result not explicitly listed.

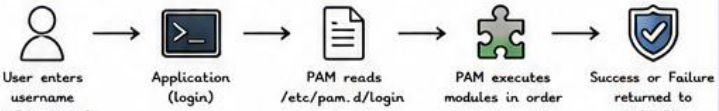
Order of modules in the file is important.

EXAMPLE: /etc/pam.d/login (Simplified)

```
# type      control      module-path                module-arguments
auth        required     pam_env.so
auth        requisite    pam_faillock.so preauth
auth        sufficient   pam_unix.so  try_first_pass nullok
auth        required     pam_deny.so
account     required     pam_unix.so
account     required     pam_faillock.so
password    sufficient   pam_unix.so sha512 shadow nullok try_first_pass use_authtok
password    required     pam_deny.so
session     required     pam_limits.so
session     required     pam_unix.so
session     optional     pam_lastlog.so showfailed
session     required     pam_env.so  readenv=1
```

This is just an example. Actual files vary by distribution.

AUTHENTICATION FLOW (LOGIN EXAMPLE)



BEST PRACTICES

- ✓ Use strong password policies.
- ✓ Limit failed login attempts (pam_faillock).
- ✓ Disable root login over SSH.
- ✓ Use key-based authentication.
- ✓ Keep PAM configuration consistent.
- ✓ Review and test changes carefully.
- ✓ Monitor authentication logs regularly.

IMPORTANT FILES

- /etc/passwd – User accounts
- /etc/shadow – Encrypted passwords
- /etc/group – Groups
- /etc/gshadow – Group passwords
- /etc/security/pwquality.conf – Password quality rules
- /etc/login.defs – Login defaults

COMMON COMMANDS

```
id <user>                       Show user identity
last                            Show last logins
lastlog                         Show last login per user
faillock --user <user>          Show failed login attempts
faillock --reset -user <user>   Reset failed attempts
```

```
authconfig --list                    Show auth configuration (RHEL/CentOS)
authconfig --enableldaps --update    Enable LDAP auth (RHEL/CentOS)
authselect current                   Show current auth profile (RHEL 8+)
authselect select sssd --force       Select auth profile (RHEL 8+)
```

QUICK RECAP

- ★ Authentication verifies WHO you are.
- ★ PAM provides HOW authentication is done.
- ★ PAM uses modules defined in /etc/pam.d/<service>.
- ★ Order and control flags decide the outcome.
- ★ Proper configuration = secure and reliable authentication.

Secure Authentication. Strong Access Control. Safer Systems.

11. FIREWALL + NETWORK SECURITY

WHAT IS A FIREWALL?

- A firewall is a security system that monitors and controls incoming and outgoing network traffic based on predefined security rules.
- It acts as a barrier between trusted (internal) networks and untrusted (external) networks.
- Goal: Allow legitimate traffic, Block malicious traffic.



WHAT IS NETWORK SECURITY?

- Network security protects data and resources as they are transmitted over networks.

Focus Areas:

- ✓ Confidentiality
- ✓ Integrity
- ✓ Availability
- ✓ Authentication
- ✓ Authorization
- ✓ Non-repudiation



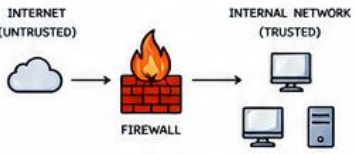
FIREWALL TYPES

Type	Description
Packet Filtering Firewall	Filters packets based on IP, port, protocol and interface.
Stateful Inspection Firewall	Tracks active connections and makes decisions based on connection state.
Application Layer Firewall	Inspects application data (e.g., HTTP, FTP, DNS).
Next-Gen Firewall (NGFW)	Inludes features like IDS/IPS, application awareness, SSL inspection.
Host-Based Firewall	Runs on individual host (e.g., Linux iptables / firewalld).

FIREWALL CONCEPTS

Concept	Description
Rule	A condition that defines what traffic is allowed or denied.
Policy	A set of rules applied to traffic.
Chain	A list of rules in netfilter/iptables (e.g., INPUT, OUTPUT).
Match	Criteria used to match packets (IP, port, protocol, interface).
Action	What to do with matched packets: ACCEPT, DROP, REJECT, LOG.
Default Policy	Action applied to packets that do not match any rule.

FIREWALL ARCHITECTURE



DUTIES OF FIREWALL

- ✓ Allow authorized traffic
- ✓ Block unauthorized traffic
- ✓ Log and monitor activity
- ✓ Protect internal resources

NETWORK SECURITY GOALS

Confidentiality
Ensure data is accessible only to authorized users.

Integrity
Ensure data is accurate and not altered.

Availability
Ensure resources are available when needed.

Authentication
Verify the identity of users/devices.

Authorization
Determine what an authenticated user is allowed to do.

LINUX FIREWALL TOOLS

Tool	Description	Used In
iptables	User-space utility for configuring netfilter firewall rules.	All major Linux distros (legacy)
nftables	Modern framework to manage packet filtering rules.	Modern Linux distros
firewalld	Frontend for nftables, easy management with zones.	RHEL, CentOS, Fedora
ufw	Uncomplicated Firewall, simple interface for iptables.	Ubuntu, Debian
fail2ban	Blocks IPs that show malicious behavior (e.g., brute force).	All Linux distros

IPTABLES BASICS

Common Chains

- INPUT : Traffic coming to the local system
- OUTPUT : Traffic going from the local system
- FORWARD : Traffic passing through the system

Default Policy Example

```
# Flush existing rules and set default policy
iptables -F
iptables -X
iptables -P INPUT DROP
iptables -P FORWARD DROP
iptables -P OUTPUT ACCEPT
```

IPTABLES COMMON RULES (EXAMPLES)

Rule	Description
iptables -A INPUT -p tcp --dport 22 -j ACCEPT	Allow SSH
iptables -A INPUT -p tcp --dport 80 -j ACCEPT	Allow HTTP
iptables -A INPUT -p tcp --dport 443 -j ACCEPT	Allow HTTPS
iptables -A INPUT -s 192.168.1.0/24 -j ACCEPT	Allow from subnet
iptables -A INPUT -j DROP	Drop all other traffic
iptables -A INPUT -p icmp -j ACCEPT	Allow ICMP (ping)
iptables -A INPUT -m state --state ESTABLISHED, RELATED -j ACCEPT	Allow established connections

FIREWALLD (ZONES)

firewalld uses zones to manage rules.

Zone	Purpose
public	For public/untrusted networks (default)
internal	For internal/trusted networks
trusted	All traffic allowed
dmz	For demilitarized zone networks
block	Drop all incoming connections
drop	Drop all traffic (no replies)

Common Firewalld Commands

```
systemctl enable --now firewalld                  # Start & enable the service
firewall-cmd --get-active-zones                   # List active zones
firewall-cmd --permanent --add-service=ssh        # Allow SSH
firewall-cmd --permanent --add-port=8080/tcp      # Allow port
firewall-cmd --reload                             # Reload rules
firewall-cmd --list-all                           # List all rules
```

UFW (UNCOMPLICATED FIREWALL)

Simple firewall for Ubuntu/Debian.

Common UFW Commands

```
ufw enable                          # Enable firewall
ufw status                          # Check status
ufw allow 22/tcp                    # Allow SSH
ufw allow 80/tcp                    # Allow HTTP
ufw allow from 192.168.1.0/24       # Allow subnet
ufw deny 23/tcp                     # Block Telnet
ufw delete allow 22/tcp             # Remove rule
ufw reset                           # Reset all rules
```

NETWORK SECURITY BEST PRACTICES

- ✓ Use firewalls to allow only necessary traffic.
- ✓ Keep systems and packages up to date.
- ✓ Disable unused services and ports.
- ✓ Use SSH keys instead of passwords.
- ✓ Implement strong password policies.
- ✓ Use fail2ban to block brute force attempts.
- ✓ Monitor logs and set up alerts.
- ✓ Use VPN for remote access.
- ✓ Enable SELinux/AppArmor for extra protection.
- ✓ Regularly backup and test your system.

COMMON PORTS

Port	Protocol	Service
22	TCP	SSH
23	TCP	Telnet (Insecure)
53	TCP/UDP	DNS
80	TCP	HTTP
443	TCP	HTTPS
3306	TCP	MySQL
5432	TCP	PostgreSQL
6379	TCP	Redis

FAIL2BAN (BRUTE FORCE PROTECTION)

Monitors logs and bans malicious IPs automatically.

Common fail2ban Commands

```
systemctl enable --now fail2ban          # Start & enable
fail2ban-client status                   # Overall status
fail2ban-client status sshd              # SSH jail status
ipta2ban-client set sshd unbanip 1.2.3.4 # Unban IP
ipta2ban-client set sshd banip 1.2.3.4   # Ban IP
```

It works with jails like: sshd, apache, postfix, mysql-auth, etc.

CHECK NETWORK CONNECTIONS & RULES

Commands	Purpose
ss -tulnp	List listening ports and processes
netstat -tulnp	(Legacy) List listening ports
iptables -L -n -v	List iptables rules with counters
nft list ruleset	List nftables rules
ip a	Show IP addresses
ip r	Show routing table

QUICK RECAP

- Firewalls control and monitor network traffic.
- iptables/nftables are powerful firewall backends.
- firewalld and ufw provide easy management.
- Always follow security best practices.

EXAMPLE SCENARIO

Allow SSH (22), HTTP (80), HTTPS (443) and block everything elee.

```
iptables -P INPUT DROP
iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
iptables -A INPUT -p tcp --dport 22 -j ACCEPT
iptables -A INPUT -p tcp --dport 80 -j ACCEPT
iptables -A INPUT -p tcp --dport 443 -j ACCEPT
iptables -A INPUT -p icmp -j ACCEPT
```

REMEMBER

A strong firewall + good security habits
= A secure Linux system.

Secure the Network, Protect the System, Ensure the Trust.

12. LINUX HARDENING + SECURE CONFIGURATION

Linux Hardening is the process of reducing the attack surface and securing the system by applying secure configurations and best practices.

WHAT IS LINUX HARDENING?

- Securing the system, services, users and network by applying secure configurations.
- Minimizes vulnerabilities and prevents unauthorized access.
- Follows the principle of Least Privilege and Defense in Depth.

BENEFITS

- ✓ Reduces attack surface
- ✓ Protects data and systems
- ✓ Prevents unauthorized access
- ✓ Improves system stability and compliance
- ✓ Helps in passing security audits

HARDENING PRINCIPLES

1. Minimize attack surface
2. Enforce least privilege
3. Defense in depth
4. Secure by default
5. Keep it simple
6. Monitor and audit continuously
7. Keep system updated and patched

KEY AREAS TO HARDEN

- User & Authentication
- File Permissions & Ownership
- Services & Daemons
- Network & Firewall
- System Configuration (Kernel, Sysctl)
- Logging & Auditing
- Updates & Packages

USER & AUTHENTICATION HARDENING

Action	Recommendation / Example
Disable root login	PermitRootLogin no in /etc/ssh/sshd_config
Use strong password policy	Configure in /etc/login.defs and /etc/security/pwquality.conf
Password aging	Set PASS_MAX_DAYS, PASS_MIN_DAYS, PASS_WARN_AGE
Lock idle accounts	Use usermod -L <user>
Remove unnecessary users	Delete or lock inactive users
Use sudo, not root	Grant limited sudo access to users
Use SSH key authentication	Disallow passwords in SSH (PasswordAuthentication no)

FILE PERMISSIONS & OWNERSHIP

- ✓ Set correct file/directory ownership (chown user:group file)
- ✓ Apply least privilege permissions (chmod)
- ✓ Remove world-writable permissions
- ✓ Protect sensitive files

Important Files	Recommended Permissions
/etc/passwd	644
/etc/shadow	640
/etc/sudoers	440
/etc/ssh/sshd_config	600
/root	700

SERVICES & DAEMONS HARDENING

- Disable unnecessary services

```
systemctl list-unit-files --type=service
systemctl disable <service>
systemctl stop <eervice>
```

- Enable only essential services
- Set services to start only when needed
- Use systemctl mask <service> to prevent start
- Regularly review running services (systemctl --type=service)

NETWORK & FIREWALL HARDENING

- Use Firewalls (iptables / nftables / firewalld / ufw)

```
# UFW example (simple)
ufw default deny incoming
ufw default allow outgoing
ufw allow ssh
ufw enable
```

- Allow only required ports and services
- Block unnecessary incoming connections
- Disable IP forwarding if not required

```
echo "net.ipv4.ip_forward = 0" >> /etc/sysctl.conf
```

- Use fail2ban to block brute force attacks

SYSTEM CONFIGURATION (SYSCTL)

Tune kernel parameters for security
Edit /etc/sysctl.conf

```
# Disable IP spoofing
net.ipv4.conf.all.rp_filter = 1
# Do not accept ICMP redirects
net.ipv4.conf.all.accept_redirects = 0
# Ignore send redirects
net.ipv4.conf.all.send_redirects = 0
# Log suspicious packets
net.ipv4.conf.all.log_martians = 1
```

Apply changes:

```
sysctl -p
```

SECURE SSH CONFIGURATION

✓ Edit /etc/ssh/sshd_config

```
Port 22
Protocol 2
PermitRootLogin no
PasswordAuthentication no
ChallengeResponseAuthentication no
UsePAM yes
MaxAuthTries 3
LoginGraceTime 30s
AllowUsers <allowed_users>
```

✓ After changes, restart SSH service

```
systemctl restart sshd
```

LOGGING & AUDITING

✓ Enable and review logs regularly

```
journalctl -xe
journalctl -u ssh
lastlog
```

✓ Enable auditd for system call auditing

```
systemctl | enable --now auditd
```

✓ Audit important files and directories

```
auditctl -w /etc/passwd -p wa -k passwd_changes
```

✓ Monitor failed login attempts

```
grep "Failed password" /var/log/auth.log
```

UPDATES & PATCH MANAGEMENT

✓ Keep OS and packages up to date

```
# Debian/Ubuntu
apt update && apt upgrade -y

# RHEL/CentOS
yum update -y  (or) dnf update -y
```

✓ Enable automatic security updates (if possible)

✓ Remove unused packages

✓ Reboot after kernel updates

LINUX HARDENING CHECKLIST

Users & Auth	☑ Disable root login, use sudo ☑ Strong password policy ☑ Remove inactive users ☑ Use SSH keys
File Permissions	☑ Correct ownership and permissions ☑ Protect sensitive files ☑ Remove SUID/SGID where not needed
Services	☑ Disable unnecessary services ☑ Enable only required services ☑ Mask unused services
Network	☑ Enable firewall ☑ Allow only required ports ☑ Disable IP forwarding (if not needed)
SSH	☑ Change default port (optional) ☑ Disable password auth ☑ Limit login attempts
System Config	☑ Set secure sysctl parameters ☑ Disable unused filesystems and modules ☑ Set umask 027
Logging & Audit	☑ Enable auditd ☑ Monitor logs ☑ Set log rotation ☑ Alert on suspicious activity
Updates	☑ Regularly update OS and packages ☑ Remove unnecessary packages ☑ Reboot when needed

QUICK REFERENCE COMMANDS

id	Show user identity
who	Logged in users
last	Login history
ss -tulnp	List listening ports
netstat -rn	Routing table
ps aux	Running processes
ls -l /etc/passwd	Check file permissions
find / -perm -4000 -type f 2>/dev/null	Find SUID files
ufw status	Firewall status
systemctl --failed	Failed services
chkconfig --list or systemctl list-unit-files	List services

IMPORTANT FILES

File / Directory	Purpose
/etc/passwd	User account information
/etc/shadow	Encrypted passwords
/etc/group	Group information
/etc/sudoers	Sudo access configuration
/etc/ssh/sshd_config	SSH server configuration
/etc/pam.d/	PAM authentication configs
/var/log/	System log files
/etc/sysctl.conf	Kernel parameters

BEST PRACTICES

- ✓ Always follow the principle of least privilege
- ✓ Minimize installed packages and services
- ✓ Regularly review configs and logs
- ✓ Backup critical configs before changes
- ✓ Document your hardening steps
- ✓ Test changes in a non-production environment

REMEMBER

Secure configuration is not a one-time task.

It is an ongoing process of review, monitoring and improvement.

HARDEN TODAY. SECURE TOMORROW. | A LITTLE HARDENING GOES A LONG WAY!

13. LOGS + AUDITING + MONITORING

WHY IT MATTERS?

- ✓ Helps in troubleshoting and root cause analysis
- ✓ Detects security incidents and suspicious activities
- ✓ Ensures compliance and accountability
- ✓ Provides visibility into system health and performance
- ✓ Supports forensic analysis and auditing

KEY CONCEPTS

- Logs: Records of system, application, security and audit events.
- Auditing: Tracking user activities and system calls for security and compliance.
- Monitoring: Real-time observation of system & application health, performance and availability.



LOG SOURCES

- System Logs: Kernel, systemd, services
- Security Logs: Auth, sudo, ssh, PAM, audit
- Application Logs: Web servers, DB, apps
- Network Logs: Firewall, IDS/IPS, connections
- Audit Logs: User actions, commands, file access
- Performance Metrics: CPU, Memory, Disk, Network
- External Logs: Cloud, containers, applications

IMPORTANT LOG FILES

Log File	Purpose	Key Information
/var/log/syslog (or /var/log/messages)	General system log	Boot, system events, services, errors
/var/log/auth.log (or secure)	Authentication log	Login success/failure, sudo, ssh, PAM
/var/log/kern.log	Kernel messages	Kernel events, warnings, driver issues
/var/log/dmesg	Boot & hardware log	Hardware detection, boot messages
/var/log/audit/audit.log	Audit subsystem log	User actions, syscalls, security events
/var/log/secure	Security log (RHEL/CentOS)	Similar to auth.log
/var/log/faillog	Failed login attempts	Failed logins per user
/var/log/lastlog	Last login info	Last login time per user
/var/log/wtmp	Login/logout records	All logins and logouts
/var/log/cron	Cron job logs	Cron job execution
/var/log/mail.log	Mail server logs	Email server events
/var/log/nginx/access.log	Web server access	HTTP access requests
/var/log/nginx/error.log	Web server errors	Nginx errors

Log file locations may vary by distribution and service configuration.

VIEWING & FILTERING LOGS

```
# View entire log
cat /var/log/syslog

# View last 50 lines
tail -n 50 /var/log/auth.log

# Follow log in real-time
tail -f /var/log/syslog

# Show logs for today
journalctl --since "today"

# Filter logs by time
journalctl --since "2025-05-15 10:00" \
           --until "2025-05-15 12:00"

# Filter by service
journalctl -u sshd

# Filter by priority (error and above)
journalctl -p err

# Search keyword in logs
grep "failed password" /var/log/auth.log

# Count failed login attempts
grep "Failed password" /var/log/auth.log |wc -l

# Show last logins
last

# Show who is logged in
who
```

AUDITING (LINUX AUDIT SYSTEM)

Linux Audit (auditd) tracks system calls, file access, user actions for security and compliance.

Check Status

```
systemctl status auditd
```

View Audit Logs

```
ausearch -m USER_LOGIN      # User login events
ausearch -m EXECVE          # Executed commands
ausearch -f /etc/passwd     # Access to passwd file
auditctl -l                 # List active audit rules
```

Add Audit Rule (Example)

```
auditctl -w /etc/passswd -p wa -k passwd_changes
# Watch file for write/attribute changes

auditctl -a always,exit -F arch=b64 -S execve -k exec
# Monitor command execution
```

Audit Logs Location

```
/var/log/audit/audit.log
```

MONITORING SYSTEM HEALTH

CPU & Load	top or htop
Memory Usage	free -h
Disk Usage	df -h
Disk I/O	iostat -xz 1 5
Network Connections	ss -tulnp
Process List	ps aux
System Uptime	uptime

REAL-TIME LOG MONITORING TOOLS

Tool	Purpose
journalctl	Query and monitor systemd journals
lnav	Advanced log file viewer (CLI) Install: sudo apt install lnav (Debian/Ubuntu) sudo yum install lnav (RHEL/CentOS)
multitail	Monitor multiple log files in real-time multitail /var/log/syslog /var/log/auth.log
logwatch	Generate daily/weekly log reports sudo logwatch --output mail --mailto admin@example.com
goaccess	Real-time web log analyzer (Nginx/Apache) goaccess /var/log/nginx/access.log
fail2ban	Monitors logs and bans malicious IPs e.g., protects against brute-force attacks

COMMON MONITORING TOOLS

Tool	Purpose	Example
prometheus	Metrics collection & alerting	Collects system/application metrics
grafana	Visualization & dashboards	Dashboards for metrics visualization
netdata	Real-time system monitoring	Lightweight real-time monitoring
zabbix	Enterprise monitoring solution	Servers, networks, apps monitoring
nagios	Infrastructure monitoring	Alerts & monitoring plugins
uptime kuma	Uptime monitoring (HTTP, TCP, etc.)	Monitor website/service availability

LOG ROTATION (LOGROTATE)

Prevents log files from growing indefinitely.

Config file: /etc/logrotate.conf
Additional configs: /etc/logrotate.d/*

Example /etc/logrotate.d/syslog

```
/var/log/syslog {
 daily
 missingok
 rotate 7
 compress
 delaycompress
 notifempty
 create 0640 syslog adm
 postrotate
    systemctl reload rsyslog 2>/dev/null  \
    || systemctl reload syslog 2>/dev/null  \
    || true
 endscript
}
```

AUDIT & MONITORING BEST PRACTICES

- ✓ Centralize logs (e.g., ELK/EFK Stack: Elasticsearch, Logstash, Kibana).
- ✓ Enable and configure auditd for critical systems.
- ✓ Monitor failed login attempts and set up alerts.
- ✓ Keep system and applications updated.
- ✓ Set up log rotation and retention policies.
- ✓ Use least privilege and monitor privilege escalation.
- ✓ Regularly review logs and audit trails.
- ✓ Automate alerts for critical events.
- ✓ Backup important logs securely.



EXAMPLE: DETECT BRUTE FORCE ATTACKS

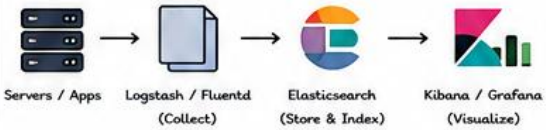
```
# Find IPs with more than 10 failed login attempts
in the last 5 minutes

grep "Failed password" /var/log/auth.log \
    | grep -v "pam_unix(sudo)" \
    | awk '{print $11}' \
    | sort | uniq -c | sort -nr | head

# Block IP using iptables (example)
iptables -A INPUT -s <IP> -j DROP

# Or use fail2ban (recommended)
systemctl enable --now fail2ban
```

CENTRALIZED LOGGING STACK (EXAMPLE)



Centralized, searchable, and visual logs with alerts.

QUICK REFERENCE COMMANDS

Command	Purpose
journalctl -xe	Show recent logs (with errors)
journalctl -b -1	Show logs from previous boot
systemctl status <service>	Check service status
dmesg | less	View kernel ring buffer
grep -i error /var/log/syslog	Search errors in syslog
watch -n 1 "df -h"	Monitor disk usage live

REMEMBER

You can't protect what you can't see.

Logs tell you what happened.

Auditing shows who did it.

Monitoring tells you what's happening now.

Stay visible, stay secure!

LOG IT. AUDIT IT. MONITOR IT. PROTECT IT.

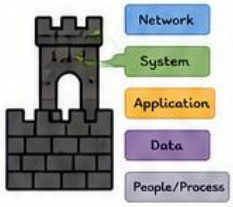
14. COMMON LINUX SECURITY THREATS + DEFENSE

WHY IT MATTERS?

- Linux is secure by design, but misconfiguration, weak practices and unpatched systems can lead to compromise.
- Understanding common threats helps you protect systems and data effectively.

KEY PRINCIPLE: DEFENSE IN DEPTH

- ✓ No single control is perfect.
- ✓ Use multiple layers of security.
- ✓ Prevention, Detection, Response and Recovery.



ATTACK SURFACE

- Open ports & services
- Unpatched software
- Weak authentication
- Misconfigurations
- Insecure file permissions
- Exposed credentials
- Untrusted network access



COMMON LINUX SECURITY THREATS + DEFENSE

#	THREAT	DESCRIPTION / EXAMPLE	IMPACT	DEFENSE / MITIGATION
1	Brute Force Attacks (SSH/Passwords)	Attacker tries many passwords to gain access via SSH or other services.	Unauthorized access, privilege escalation.	• Use strong passwords & SSH keys • Disable root login (PermitRootLogin no) • Fail2ban / rate limiting / account lockout
2	Malware & Backdoors	Malicious software or backdoors installed to maintain hidden access.	Data theft, system takeover, persistence.	• Keep system updated • Use malware scanner (ClamAV, rkhunter) • Monitor file integrity (AIDE, Tripwire)
3	Privilege Escalation	Attacker gains higher privileges from a low-privileged account.	Full system compromise.	• Apply least privilege • Keep kernel & software updated • Audit sudoers and SUID/SGID files
4	Unpatched Vulnerabilities	Known vulnerabilities in OS, kernel or applications not patched.	Remote code execution, data breach.	• Regular updates and security patches • Subscribe to security advisories • Automated patch management
5	Weak Authentication	Weak passwords, reused passwords, or default accounts.	Unauthorized access.	• Enforce strong password policy • Disable unused/default accounts • Enable MFA where possible
6	Open Ports & Unnecessary Services	Unneeded services or open ports expose system to attacks.	Service exploitation, network intrusion.	• Close unused ports (iptables / firewalld) • Disable unnecessary services • Use firewall and port filtering
7	Misconfigurations	Incorrect system or application configurations.	Information disclosure, service disruption.	• Follow secure configuration baselines • Disable unused features • Regular security reviews
8	Credential Theft	Stealing passwords, SSH keys or tokens from system or memory.	Account takeover, lateral movement.	• Protect credentials & keys • Use SSH key passphrases • Monitor for suspicious activity
9	DoS / DDoS Attacks	Flooding services/network to make them unavailable.	Service downtime, revenue loss.	• Rate limiting / connection limits • Use firewalls, IDS/IPS • Leverage CDN / DDoS protection
10	Web Exploits (Web Servers)	Attacks like SQLi, XSS, RCE on web applications running on Linux.	Data breach, defacement, server takeover.	• Keep web stack updated • Use WAF (Web Application Firewall) • Input validation, least privilege
11	Insider Threats	Malicious or careless actions by authorized users.	Data leakage, sabotage.	• Least privilege access • Monitor and audit user activity • Strong accountability and logging
12	Physical Access	Unauthorized physical access to servers or devices.	Data theft, hardware manipulation.	• Secure server rooms • Use BIOS/boot passwords • Encrypt disks (LUKS)

COMMON ATTACK VECTORS

- Phishing & social engineering
- Exploiting software vulnerabilities
- Weak or stolen credentials
- Misconfigured network/services
- Removable media / physical access

DETECTION & MONITORING

- ✓ Monitor logs: /var/log/*
- ✓ Use auditd for system call auditing
- ✓ Monitor file integrity (AIDE, Tripwire)
- ✓ Use IDS/IPS (Snort, OSSEC, Suricata)
- ✓ Monitor resource usage & processes
- ✓ Set up alerts for suspicious activity

LOGS TO CHECK

/var/log/auth.log	- Authentication logs
/var/log/secure	- Security events (RHEL/CentOS)
/var/log/syslog	- General system logs
/var/log/kern.log	- Kernel messages
/var/log/dmesg	- Boot & hardware logs
/var/log/audit/audit.log	- Audit logs
/var/log/cron	- Cron job logs
/var/log/httpd/	- Web server logs

KEY DEFENSE STRATEGIES

- ✓ Keep systems & software updated
- ✓ Enforce strong authentication
- ✓ Apply least privilege access
- ✓ Harden system configurations
- ✓ Use firewalls & network security
- ✓ Monitor logs & set alerts
- ✓ Backup & disaster recovery
- ✓ Regular security assessments

SYSTEM HARDENING QUICK WINS

- ✓ Disable root login: PermitRootLogin no
- ✓ Use SSH keys instead of passwords
- ✓ Change default ports (e.g., SSH 22)
- ✓ Disable unused services
- ✓ Set proper file permissions
- ✓ Enable firewall (ufw/firewalld)
- ✓ Enable SELinux/AppArmor
- ✓ Regularly review users and groups

FIREWALL BASICS (EXAMPLE)

Using UFW (Uncomplicated Firewall)

```
# Allow only required services
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow OpenSSH
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw enable
sudo ufw status verbose
```

AUDITD BASICS (EXAMPLE)

Track important events

```
# Install auditd
sudo yum install audit -y      # RHEL/CentOS
sudo apt install auditd -y     # Debian/Ubuntu
# Enable and start
sudo systemctl enable auditd
sudo systemctl start auditd
# Add rules (example: watch /etc/passwd)
sudo auditctl -w /etc/passwd -p wa -k passwd_changes
```

INCIDENT RESPONSE (BASIC STEPS)

1. Identify the incident
2. Contain the threat
3. Eradicate the cause
4. Recover the system
5. Review and strengthen defenses

REMEMBER

Security is not a product,
it's a continuous process.
Stay alert. Stay updated. Stay secure.

QUICK REFERENCE COMMANDS

```
who                                  # Show logged in users
last                                 # Login history
ss -tulnp                            # List open ports
ps aux                               # Running processes
sudo lsof -i -P -n                   # Open network connections
grep "Failed password" /var/log/auth.log  # Failed login attempts
sudo rkhunter --check                # Rootkit check
sudo chkrootkit                      # Rootkit scanner
sudo lynis audit system              # Security audit
sudo fail2ban-client status          # Fail2ban status
sudo logwatch                        # Log summary report
```

USEFUL TOOLS

rkhunter	- Rootkit scanner
chkrootkit	- Rootkit checker
lynis	- Security auditing
aide	- File integrity checker
fail2ban	- Brute force protection
nmap	- Network scanner

SECURE TODAY, PROTECT TOMORROW!

15. LINUX SECURITY QUICK REVISION + INTERVIEW NOTES

Secure by default. Verify always. Monitor continuously. Least privilege everywhere.

1. ESSENTIAL SECURITY PRINCIPLES

- ✓ Least Privilege – give only required access
- ✓ Defense in Depth – multiple layers of security
- ✓ Secure by Default – enable secure settings
- ✓ Fail Securely – system should resist attacks
- ✓ Keep Updated – patch OS, apps, and configs
- ✓ Monitor & Audit – detect, respond, improve
- ✓ Backup & Recovery – prepare for the worst

2. SECURITY CHECKLIST (QUICK REVIEW)

Users & Auth	Strong passwords, SSH keys, disable root login
File Permissions	Least privilege, correct ownership, remove SUID/SGID
Services	Disable unused services, run only what's needed
Network	Firewall enabled, only required ports open
SSH	Key auth, non-default port (optional), fail2ban
System	Harden kernel, disable unused modules
Logging & Audit	Logs enabled, auditd active, log rotation
Updates	Keep OS & packages updated

3. HARDENING QUICK RECAP

- Disable root login: `PermitRootLogin no`
- Use sudo, not root
- Strong password policy (length, complexity, expiry)
- Disable unused users and lock accounts
- Enable firewall (ufw / firewalld)
- Enable SELinux/AppArmor
- Disable IPv6 if not used `net.ipv6.conf.all.disable_ipv6 = 1`
- Secure SSH: keys, no passwords, limit attempts
- Disable unused filesystems and services
- Regular backups and test restore

4. COMMON THREATS & DEFENSE (AT A GLANCE)

Threat	Example	Impact	Defense
Brute Force (SSH/Passwords)	Multiple password login attempts	Unauthorized access	Strong passwords, key auth, Fail2ban, rate limiting
Malware / Backdoors	Malicious software or hidden access	Data theft, persistence	Keep updated, AV scan, file integrity monitoring
Privilege Escalation	Exploit to gain higher privileges	Full system compromise	Least privilege, patching, secure configs
Unpatched Vulnerabilities	Known bug not patched	Remote code execution	Regular updates, security advisories
Weak Authentication	Weak/ reused passwords	Unauthorized access	Strong policy, MFA where possible, lockout
Open Ports / Services	Exposed services or ports	Service exploitation, network intrusion	Close unused ports, firewall rules
Misconfigurations	Insecure system settings	Info disclosure, compromise	Follow best practices, security baselines
Credential Theft	Steal passwords, keys, tokens	Account takeover	Protect credentials, SSH key passphrases, monitoring
DoS / DDoS	Flood services/ network	Service downtime	Rate limiting, firewall, IDS/IPS, CDN
Web Exploits	SQLi, XSS, RCE on web apps	Data breach, defacement	WAF, input validation, least privilege
Insider Threats	Malicious or careless insider	Data leakage, sabotage	Least privilege, monitoring, auditing
Physical Access	Server or device access	Theft, tampering	Lock racks, BIOS passwords, disk encryption

5. IMPORTANT FILES & LOCATIONS

`/etc/passwd`	User account info
`/etc/shadow`	Encrypted passwords
`/etc/group`	Group info
`/etc/sudoers`	Sudo permissions
`/etc/ssh/sshd_config`	SSH server config
`/etc/hosts`	Host name resolution
`/etc/fstab`	File systems
`/var/log/`	All system logs
`/var/log/auth.log`	Authentication logs
`/var/log/secure`	Security events (RHEL)
`/var/log/syslog`	General system logs
`/var/log/audit/audit.log`	Audit logs
`/etc/selinux/config`	SELinux config
`/etc/apparmor/`	AppArmor profiles
`/etc/cron.*`	Scheduled tasks
`~/.ssh/authorized_keys`	SSH authorized keys

6. USEFUL COMMANDS (CHEAT SHEET)

`who`	Show logged in users
`last`	Login history
`ss -tulpn`	List listening ports
`netstat -tulpn`	List listening ports (legacy)
`ps aux`	Running processes
`top / htop`	System resource usage
`df -h`	Disk usage
`free -h`	Memory usage
`systemctl status <svc>`	Service status
`journalctl -xe`	View system logs
`grep "Failed password" /var/log/auth.log`	Failed login attempts
`sudo ausearch -m USER_LOGIN`	Audit: user login events
`ufw status`	Firewall status
`sudo ufw allow 22/tcp`	Allow SSH in UFW
`sudo firewall-cmd --list-all`	Firewalld status
`find / -perm -4000 -type f 2>/dev/null`	Find SUID files
`rpm -Va | less`	Verify RPM packages
`chkrootkit`	Rootkit checker
`rkhunter --check`	Rootkit scanner

7. LOGS QUICK RECAP

`/var/log/messages`	General system messages
`/var/log/auth.log` (RHEL/CentOS)	Auth events (login, sudo, ssh)
`/var/log/secure` (Debian/Ubuntu)	Security events
`/var/log/kern.log`	Kernel messages
`/var/log/dmesg`	Boot & hardware messages
`/var/log/syslog`	General logs
`/var/log/audit/audit.log`	Audit logs

8. BEST PRACTICES (MUST DO)

- ✓ Keep OS, kernel & packages updated
- ✓ Use strong passwords & SSH keys
- ✓ Enable firewall, close unused ports
- ✓ Run services as non-root users
- ✓ Disable unused services & users
- ✓ Enable audit logging
- ✓ Monitor logs & set alerts
- ✓ Backup important data regularly
- ✓ Use encryption for data in transit and at rest
- ✓ Regular security assessments

9. INTERVIEW NOTES – COMMON QUESTIONS

Q1. How do you secure SSH?
: Disable root login, use key authentication, change default port (optional), limit login attempts (Fail2ban), allow only specific users/IPs, keep SSH updated.

Q2. How do you check if a system was compromised?
: Check logs (`/var/log/*`, audit logs), look for strange users, recent changes in cron, SUID files, running processes, network connections, file integrity. Use tools: `chkrootkit`, `rkhunter`, `auditd`.

Q3. How do you prevent brute force attacks?
: Strong passwords, SSH keys, Fail2ban, account lockout, rate limiting, firewall rules to block suspicious IPs.

Q4. What is least privilege?
: Give users/processes only the minimum permissions required to perform their tasks.

Q5. How do you harden a Linux server?
: Update system, disable unused services, secure SSH, enable firewall, configure SELinux/AppArmor, set strong passwords, enable audit logs, monitor and review logs.

Q6. How do you analyze logs?
: Use `journalctl`, `grep`, `ausearch`, `tail`, `less`. Look for failed logins, privilege escalations, sudo usage, service failures, unusual activity.

10. SECURITY CONFIG SNIPPETS (QUICK REFERENCE)

SSH – `/etc/ssh/sshd_config` (Key Settings)

```
PermitRootLogin no
PasswordAuthentication no
PubkeyAuthentication yes
AllowUsers user1 user2
MaxAuthTries 3
LoginGraceTime 30s
Port 2222    # optional
```

Password Policy – `/etc/login.defs`

```
PASS_MAX_DAYS   60
PASS_MIN_DAYS   7
PASS_WARN_AGE   7
PASS_MIN_LEN    12
```

Firewall (UFW)

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 22/ctp
sudo ufw enable
```

Sysctl Hardening – `/etc/sysctl.conf` (Example)

```
net.ipv4.ip_forward = 0
net.ipv4.conf.all.rp_filter = 1
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.all.send_redirects = 0
net.ipv6.conf.all.disable_ipv6 = 1
```

11. QUICK SECURITY AUDIT STEPS

1. Check system & kernel version
   `uname -a`
2. Check for updates
   `sudo apt update & sudo apt upgrade -y`
3. Check listening ports
   `ss -tulpn`
4. Check failed login attempts
   `grep "Failed password" /var/log/auth.log`
5. Check SUID/SGID files
   `find / -perm -4000 -type f 2>/dev/null`
6. Check running services
   `systemctl list-units --type=service --state=running`
7. Review logs & audit
   `journalctl -xe, ausearch -m USER_LOGIN`
8. Verify firewall
   `ufw status` or `firewall-cmd --list-all`
9. Check file permissions & ownership
   `ls -l /etc/passwd /etc/shadow`
10. Document & remediate findings

12. RED FLAGS (WHAT TO LOOK FOR)

- ✗ Root login enabled over SSH
- ✗ Password authentication enabled for SSH
- ✗ Default passwords or weak passwords
- ✗ Unnecessary open ports and services
- ✗ SUID files that should not be SUID
- ✗ Users with UID 0 other than root
- ✗ World-writable files or directories
- ✗ Outdated kernel or packages
- ✗ Audit service not running
- ✗ Firewall disabled or misconfigured
- ✗ Unusual processes or network connections
- ✗ Large number of failed login attempts

13. REMEMBER

Security is not a product, it's a process.

Stay alert. Stay updated. Stay secure.

14. ONE-LINER SUMMARY

Secure the OS, control access, monitor everything, and keep improving.

PLAN SECURELY. CONFIGURE SECURELY. MONITOR CONTINUOUSLY. RESPOND QUICKLY.